MAHATMA GANDHI UNIVERSITY, KOTTAYAM
B.Sc. (Hons) CHEMISTRY – Semester IV

Model Question Paper
Physical Chemistry- 1 (MG4DSCCHE201)

Time 90 mints Maximum marks 50

PART A – Multiple Choice Questions

Answer all questions. Each question carries 1 mark

1. Deviation of real gases from ideal behaviour is mainly due to
   a) Elastic collisions b) Absence of intermolecular forces
   c) Molecular volume and intermolecular attractions d) High temperature

(CO, 1 Understand)

2. Boyle temperature is the temperature at which a real gas
   a) Liquefies b) Shows maximum deviation
   c) Obeys ideal gas law over a wide pressure range d) Has zero volume

(CO1, Remember)

3. Maxwell–Boltzmann distribution law explains the distribution of
   a) Molecular energies only b) Molecular velocities
   c) Molecular collisions d) Molecular volumes

(CO 1, Remember)

4. According to Bronsted–Lowry concept, an acid is a substance that
   a) Accepts electrons b) Donates electrons
   c) Donates protons d) Accepts protons

(CO2, Understand)

5. Ostwald's dilution law is applicable to
   a) Strong electrolytes b) Weak electrolytes
   c) Non-electrolytes d) Neutral salts

(CO2, Understand)

6. The ionic product of water at 25 °C is
   a) 10^{-7} b) 10^{-14}
   c) 10^{-5} d) 10^{-2}

(CO2, Apply)

7. What is the degrees of freedom if a one component system exists in three different phases simultaneously?

a) 0 b) 1
c) 2 d) 3

(CO3, Evaluate)

8. Pattinson's process is used in the metallurgy of
   a) Copper b) Zinc
   c) Lead d) Silver

(CO3, Understand)

9. Henry's law relates
   a) Vapour pressure and mole fraction b) Solubility of a gas and pressure
   c) Osmotic pressure and concentration d) Freezing point and molar mass

(CO4, Understand)

10. Lowering of vapour pressure is a
    a) Additive property b) Constitutive property
    c) Colligative property d) Intensive property

(CO4, Understand)

10 X 1 =10

PART B – Short Answer Questions

Answer any FOUR. Each question carries 5 mark

11. Explain critical constants and describe their calculation from van der Waals equation. **(CO1, Remember)**

12. Define collision diameter, collision number, and collision frequency. **(CO1, Understand)**

13. Draw and explain the Pb-Ag two component phase diagram. Calculate the degrees of freedom **(CO 3, Analyse)**

14. Explain the mechanism of buffer action. **(CO 2, Understand)**

15. Calculate the mass of compound (molar mass =256 g/mol) to be dissolved in 75 g of benzene to lower its freezing point by 0.48 K. (Kf=5.12 K kg mol-1). **(CO 4, Apply)**

16. Calculate the pH and pOH of 0.03 M aqueous solution of HCl at 25 °C **(CO 2, Apply)**

4 X 5 =20

PART C – Short Essay Questions

Answer any TWO. Each question carries 10 mark

17. Explain the virial equation of state and express van der Waals equation in virial form. **(CO 1, Analyse)**

18. Explain mean free path of molecules and derive its relation with coefficient of viscosity. **(CO1, Apply)**

19. Explain Henderson–Hasselbalch equation and discuss hydrolysis of salts**. (CO 2, Understand)**

20. Explain abnormal molecular masses and discuss van't Hoff factor, degree of association and dissociation. **(CO4, Analyse)**

2 × 10 = 20 marks